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Note on AI Use

Part 1 GPT-5.3-Codex in Copilot

I used this AI assistant to help me with the technical side of my research. It acted like a partner that helped me think about my research question and finish my final report. My main goal was to find out ifpolitical violence in Afghanistan led to more violence in neighboring countries during the following month. The assistant helped me with coding and organizing my workflow.

When I was cleaning the data, the AI helped me find a mistake where an Excel file was incorrectly labeled as a CSV file. It helped me write the code to fix this and set up a system to clean the data automatically. This included matching month names and filtering the right countries which helped me avoid making manual mistakes.

I also used the assistant to create many charts for the project. It helped me make time series plots and heatmaps to show the data. Later it helped me build interactive charts for my web report so the results would be easier to understand.

For the statistical analysis part, the AI helped me run regression models for each country. It produced the important numbers like p-values and coefficients and put them into tables for my report. It also helped me write the HTML code for my final website. I could tell it how I wanted the layout to look and it would update the code for me. Finally, the AI helped me check my writing to make sure the tone was right and helped me organize my files for the final submission.

What's more, to optimize the rendering performance and reduce the file size of the interactive HTML report, I utilized an AI coding assistant to serialize the Plotly data arrays into Base64 encoded binary strings (bdata with specified dtypes). This front-end optimization prevents the browser from loading massive plaintext numeric arrays, making the dashboard load significantly faster.

Part 2 Gemini-3.1-Pro

I used Gemini to help me with the research design and the logic of my paper. In the beginning, I asked the AI what makes a research question important and well-defined. It suggested that I look at specific details in the ACLED database. Because of this advice, I decided to study the relationship between violence in Afghanistan and its six neighbors.

When I looked at my results, I found a negative correlation that I did not expect. The AI helped me understand this by suggesting political science theories like state capacity. This helped me explain why the data looked that way.

The AI also helped me improve the structure of my report. It looked at professional reports and suggested that I should focus more on evidence instead of just telling a story. It helped me use the correct terms for different types of events like riots or strategic developments.

In the final stage, the AI translated my draft into English and made sure I followed the rules for APA citations. It even helped me grade my own report against a rubric to see where I could improve. It gave me advice on how to make better charts and how to order my paragraphs so the argument was easy to follow. By using the AI for these tasks, I could spend more time thinking about the political analysis of the spillover effects.

Table 1. AI Roles in Different Stages

Project Phase	AI Support Provided
Thinking and Planning	Narrowing down the research question and finding theories
Data Processing	Fixing Excel errors and organizing the time series data
Statistics and Charts	Running the regression models and making interactive plots
Final Writing	Changing the tone and making sure citations are correct

Figure 1. The Project Workflow
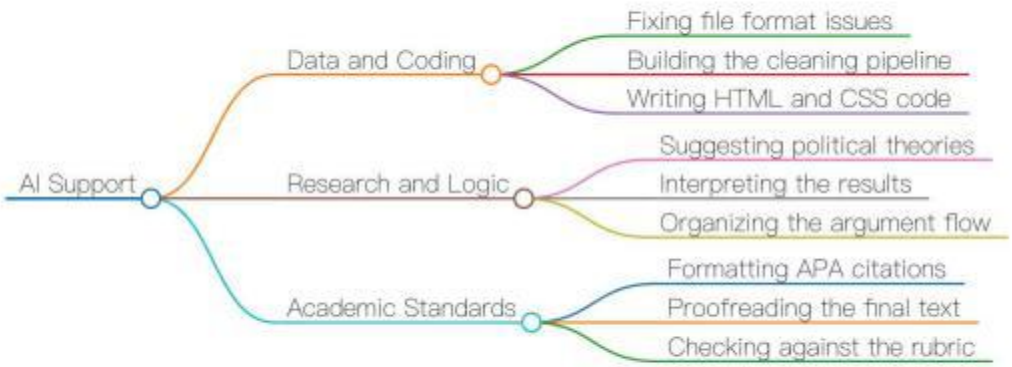
Figure 2. Specific AI Contributions